Python: warm-up exercises on branching

This notebook was generated in **pseudocode** mode.

Exercise: Check if Number is Even

Exercise Description

Write a piece of code that checks whether a number `num` is even. If so, print a message.

Your solution

```
num = 42
# step 1: use the modulo operator % to check if num is divisible by 2
# step 2: if the remainder is 0, the number is even
# step 3: print a message informing that the number is even
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Even or Odd Message

Exercise Description

Write a piece of code that checks whether a number `num` is even. Print a message about the outcome (odd or even).

Your solution

```
num = 42
# step 1: use the modulo operator % to check if num is divisible by 2
# step 2: if the remainder is 0, the number is even, print even message
# step 3: otherwise, the number is odd, print odd message
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Conditional assignment

Exercise Description

Write a piece of code that checks whether a number `num` is odd or even. Assign a proper value (e.g., 'odd' or 'even') to a variable `result`, depending on the outcome of the condition Print the outcome at the end.

Your solution

```
num = 42
# step 1: create a variable to store the result (e.g., result = "")
# step 2: use the modulo operator % to check if num is divisible by 2
# step 3: if even, assign "even" to the result variable
# step 4: if odd, assign "odd" to the result variable
# step 5: print the result
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Multiple Divisibility Check

Exercise Description

Write a piece of code that checks whether a number `num` is divisible by 2, 3, 5 and 7. Print a message when `num` is divisible by one of this number.

Your solution

```
num = 42
# step 1: check if num is divisible by 2 using modulo operator, print message if true
# step 2: check if num is divisible by 3 using modulo operator, print message if true
# step 3: check if num is divisible by 5 using modulo operator, print message if true
# step 4: check if num is divisible by 7 using modulo operator, print message if true
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```python
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Nested Conditional Logic

Exercise Description

Write a piece of code that checks whether a number `num` is odd or even. Print a message informing the user. If odd, check whether the number is divisible by 5. Print a message informing the user. If even, check whether the number is divisible by 3. Print a message informing the user.

Your solution

```python
num = 15
# step 1: check if num is even using modulo operator
# step 2: if even, print even message and check divisibility by 3
# step 3: if odd, print odd message and check divisibility by 5
# step 4: for each nested check, print appropriate message about divisibility
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise: Dog Age Calculator

Exercise Description

Write a piece of code that asks the user to insert the dog's age in human's years and calculates a dog's age in dog's years. For the first two years, a dog year is equal to 10.5 human years. After that, each dog year equals 4 human years.

Your solution

```
# step 1: ask the user for the dog's age using input() and convert to integer
# step 2: check if the age is 2 or less
# step 3: if age <= 2, multiply age by 10.5 to get dog years
# step 4: if age > 2, calculate as 2 * 10.5 + (age - 2) * 4
# step 5: print the result
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**